Optimizing Brownian heat engine with shortcut strategy

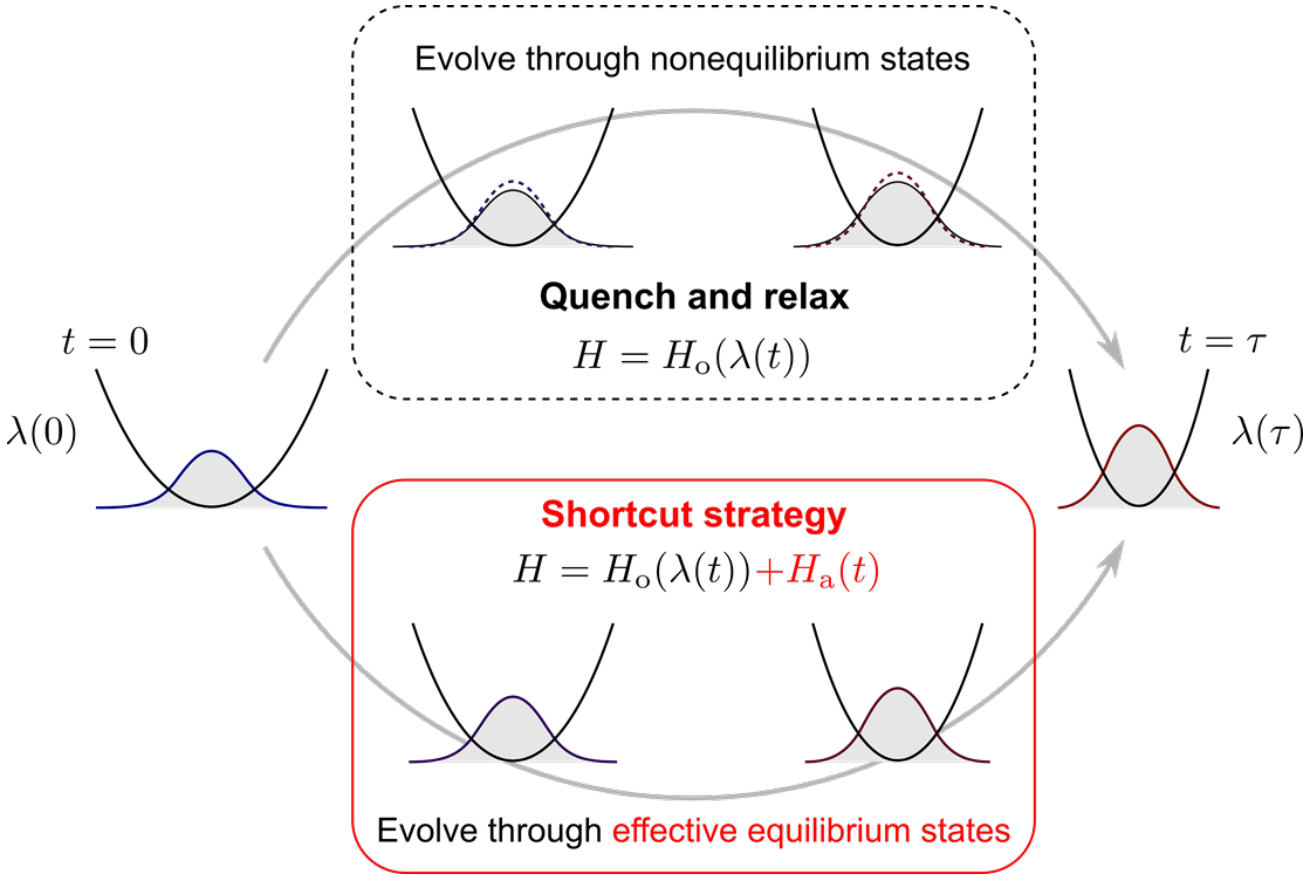
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Transformation between two equilibrium states



I. A. Martínez, A. Petrosyan, D. Guéry-Odelin, E. Trizac, and S. Ciliberto, Nat. Phys. 12, 843 (2016).
G. Li, H. T. Quan and Z. C. Tu, Phys. Rev. E 96, 012144 (2017).

Outline

- Shortcut strategy for underdamped Brownian motion
- Optimizing Brownian heat engine with shortcut strategy
- Discussions on underdamped and overdamped limits
- Summary and outlook

Shortcut strategy of evolution

- Consider a Brownian particle trapped in a potential.
- Goal: let the system evolve along

$$\rho(t) = \rho_{\mathrm{ieq}}(t) = \frac{1}{Z(\beta(t))} e^{-\beta(t) H_{\mathrm{o}}(\lambda(t))}$$

- Total Hamiltonian: $H = H_{\mathrm{o}}(\lambda(t)) + H_{\mathrm{a}}(t)$

origin auxiliary

- Dynamics: the Fokker-Planck equation

$$\frac{\partial \rho}{\partial t} = \mathscr{L}[\rho] + \mathscr{D}[\rho],$$

Deterministic $$\mathscr{L}[\rho] = -\frac{\partial}{\partial x}\left(\frac{\partial H}{\partial p}\rho\right) + \frac{\partial}{\partial p}\left(\frac{\partial H}{\partial x}\rho\right),$$

Dissipative $$\mathscr{D}[\rho] = \kappa m \frac{\partial}{\partial p}\left(\frac{\partial H}{\partial p}\rho + \frac{1}{\beta}\frac{\partial \rho}{\partial p}\right).$$

Shortcuts to isothermality: G. Li, H. T. Quan and Z. C. Tu, Phys. Rev. E 96, 012144 (2017)

Shortcut strategy of evolution

- We consider harmonic potential

$$H_{\mathrm{o}} = \frac{1}{2m}p^2 + U_{\mathrm{o}}, \qquad U_{\mathrm{o}} = \frac{1}{2}m\lambda^2 x^2,$$

- The auxiliary Hamiltonian is solved in the form

$$H_{\mathrm{a}}(x,p,t) = \dot{\lambda}h_\lambda(x,p,\lambda) + \dot{\beta}h_\beta(x,p,\lambda),$$

- where

$$h_\lambda = \frac{1}{2\kappa}\left[\frac{1}{\lambda m}(p-\kappa m x)^2 + m\lambda x^2\right],$$

$$h_\beta = \frac{p^2 + (p-\kappa m x)^2 + 2m^2\lambda^2 x^2}{4\kappa\beta m}.$$

Y. Jun and P. Y. Lai, Phys. Rev. Res. 3, 033130 (2021)

Energetics in shortcut processes

- The average input power and the average heat flux are

$$\dot{W} \equiv \left\langle \frac{\partial H}{\partial t} \right\rangle = \dot{W}_{\rm o} + \dot{W}_{\rm irr}, \qquad \dot{Q} \equiv \iint H \frac{\partial \rho_{\rm ieq}}{\partial t} dx dp = \dot{Q}_{\rm o} + \dot{Q}_{\rm irr},$$

- We divide work and heat into quasistatic and irreversible parts

$$\dot{W}_{\rm o} = \dot{\lambda} \left\langle \frac{\partial H_{\rm o}}{\partial \lambda} \right\rangle, \qquad \dot{Q}_{\rm o} = \iint H_{\rm o} \frac{\partial \rho_{\rm ieq}}{\partial t} dx dp,$$

$$\dot{W}_{\rm irr} = \left\langle \frac{\partial H_{\rm a}}{\partial t} \right\rangle, \qquad \dot{Q}_{\rm irr} = \iint H_{\rm a} \frac{\partial \rho_{\rm ieq}}{\partial t} dx dp.$$

- First law:

$$\frac{\partial \langle H_{\rm o} \rangle}{\partial t} = \dot{W}_{\rm o} + \dot{Q}_{\rm o}, \qquad \frac{\partial \langle H_{\rm a} \rangle}{\partial t} = \dot{W}_{\rm irr} + \dot{Q}_{\rm irr}.$$

$$W_{\rm irr} = \underset{\|}{\langle H_{\rm a} \rangle (\tau^+)} - \underset{\|}{\langle H_{\rm a} \rangle (0^-)} - Q_{\rm irr}$$

$H_{\rm a}$ is switched off initially and finally $\quad 0 \qquad\qquad 0$

- Maximizing the output work $-W_{\rm o} - W_{\rm irr}$ = maximizing $Q_{\rm irr}$

Thermodynamic length of irreversibility

- For shortcut processes, the irreversible heat is exactly

$$Q_{\rm irr} = -\int_0^\tau \begin{pmatrix} \dot\lambda & \dot\beta \end{pmatrix} \mathbf{g} \begin{pmatrix} \dot\lambda \\ \dot\beta \end{pmatrix} dt,$$

Thermodynamic length:

$$\mathcal{L} \equiv \int_0^1 \sqrt{\begin{pmatrix} \tilde\lambda'(s) & \tilde\beta'(s) \end{pmatrix} \mathbf{g} \begin{pmatrix} \tilde\lambda'(s) \\ \tilde\beta'(s) \end{pmatrix}} ds.$$

metric:

$$\mathbf{g} = \kappa m \begin{pmatrix} \left\langle \frac{\partial h_\lambda}{\partial p} \frac{\partial h_\lambda}{\partial p} \right\rangle & \left\langle \frac{\partial h_\lambda}{\partial p} \frac{\partial h_\beta}{\partial p} \right\rangle \\ \left\langle \frac{\partial h_\beta}{\partial p} \frac{\partial h_\lambda}{\partial p} \right\rangle & \left\langle \frac{\partial h_\beta}{\partial p} \frac{\partial h_\beta}{\partial p} \right\rangle \end{pmatrix}$$

- The irreversible work (or heat) is bounded by

$$W_{\rm irr} = -Q_{\rm irr} \geq \frac{\mathcal{L}^2}{\tau} > 0$$

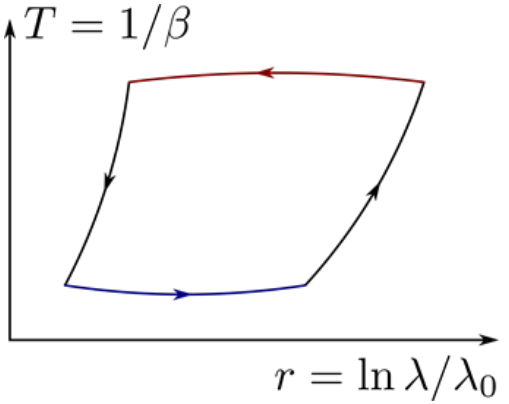
- Equality is reached for the optimal protocol.

- We next consider cyclic control of a Brownian heat engine.

G. Li, JFC, C. P. Sun, H. Dong, Phys. Rev. Lett. 128, 230603 (2022)
JFC, Phys. Rev. E 106, 054108 (2022)

Brownian heat engine: maximum power

- Consider a closed path of a heat-engine cycle $\lambda(s)$ and $\beta(s)$ in control parameter space.



- The output work is bounded by

$$-W = -W_{\rm o} - W_{\rm irr} \leq -W_{\rm o} - \mathcal{L}^2/\tau,$$

$W_{\rm o}$: work of quasistatic cycle $\mathcal{L}$: thermodynamic length of closed path

- Choose the operation time $\tau_{\rm max} = 2\mathcal{L}^2/(-W_{\rm o})$.

- The maximum power is

$$P_{\rm max} = \frac{-W}{\tau_{\rm max}} = \frac{(-W_{\rm o})^2}{4\mathcal{L}^2}.$$

JFC, Phys. Rev. E 106, 054108 (2022)

Shortcut Brownian heat engine: EMP

- Divide the path into heat absorbed and heat released paths according to $\dot{Q} > 0$ and $\dot{Q} < 0$, and define their thermodynamic lengths $\mathcal{L}_+$ and $\mathcal{L}_-$.
- Heat absorbed in a cycle $Q_+ = Q_{\mathrm{o},+} - \mathcal{L}_+\mathcal{L}/\tau.$
- Efficiency of the cycle:

$$\eta \equiv \frac{-W}{Q_+} = \frac{-W_\mathrm{o} - \mathcal{L}^2/\tau}{-W_\mathrm{o}/\eta_\mathrm{o} - \mathcal{L}_+\mathcal{L}/\tau}.$$

where the efficiency of quasistatic cycle is

$$\eta_\mathrm{o} = \frac{-W_\mathrm{o}}{Q_{\mathrm{o},+}}.$$

- Efficiency at maximum power:

$$\eta_\mathrm{EMP} = \frac{\eta_\mathrm{o}}{2 - \eta_\mathrm{o}\mathcal{L}_+/\mathcal{L}}.$$

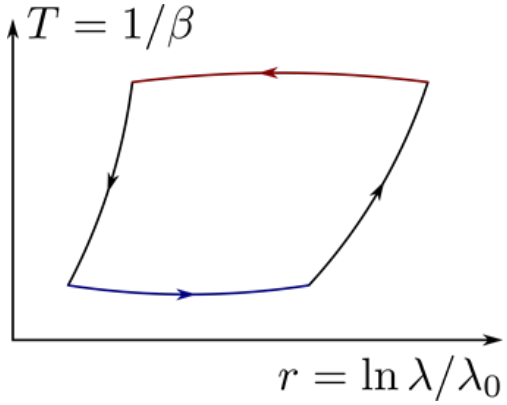
Results for harmonic potential

- Represent the control parameters as

$$\lambda = \lambda_0 e^r, \quad \beta = 1/T.$$

- Quasistatic output work of the cycle:

$$-W_{\mathrm{o}} = -\oint T dr.$$



- Proportional to the area $\mathcal{A} = \left|\oint T dr\right|$ in $r - T$ diagram.

- Thermodynamic length of the cycle

$$\mathcal{L} = \oint \sqrt{\frac{(Tdr - dT)^2}{\kappa T} + \frac{\kappa}{(\lambda_0 e^r)^2}\frac{1}{T}\left(Tdr - \frac{dT}{2}\right)^2}.$$

Dissipation in underdamped limit

$\kappa \ll \lambda$

Dissipation in overdamped limit

$\kappa \gg \lambda$

Optimization of heat-engine cycle

- For the optimal cycle on a given closed path, the power is

$$P = \frac{\mathcal{A}}{\tau} - \frac{\mathcal{L}^2}{\tau^2},$$

where $\mathcal{A}$ and $\mathcal{L}$ are the area and the thermodynamic length of the closed path.

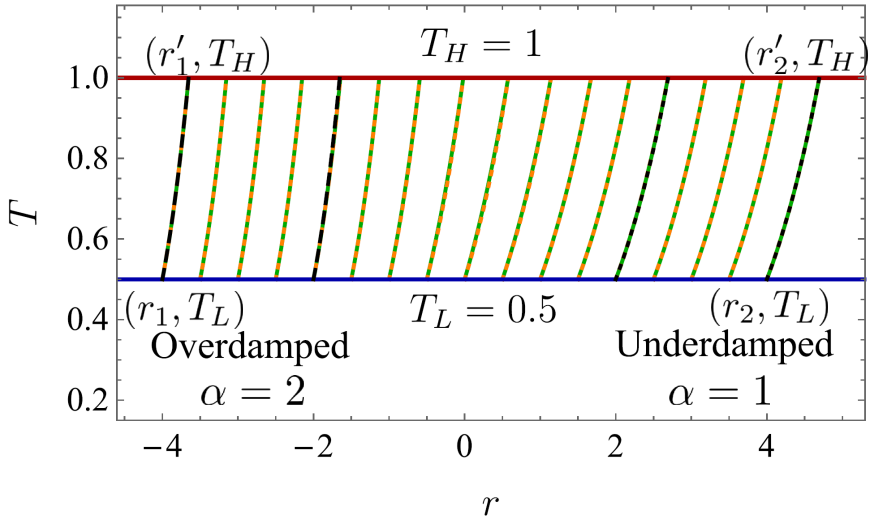
- Two aspects for optimizing cycles with shortcut strategy:
  1. Along a given closed path, manipulate the control parameters with a proper constant velocity to reach the maximum power

$$P_{\max} = \frac{\mathcal{A}^2}{4\mathcal{L}^2}.$$

  2. Find out a closed path with large $\mathcal{A}/\mathcal{L}$.

Optimization in general damped situation

- Choose two isothermal lines (T_L and T_H) and two connecting paths to form a closed cycle.



- Connecting paths are shortest geodesic curves or exponential paths

$$T(r) = T_L e^{\alpha(r-r_0)}, \qquad r_0 \le r \le r_0 + \frac{1}{\alpha}\ln(\frac{T_H}{T_L}).$$

- We choose $\alpha_{1,2}$ to minimize $\mathcal{L}_{\exp}(r_1, \alpha_1, r_2, \alpha_2)$ or fixed as $\alpha_{1,2} = 1$ or 2.

Underdamped limit

- In the underdamped limit $\kappa \ll \lambda$, the thermodynamic length is similified to

$$\mathcal{L} = \oint \sqrt{\frac{(Tdr - dT)^2}{\kappa T}}.$$

- Choose the exponential paths with $\alpha = 1$ (zero length), and obtain

$$P_{\max}^{\text{under}} = \frac{\kappa}{4}(\sqrt{T_H} - \sqrt{T_L})^2,$$

$$\eta_{\text{EMP}}^{\text{under}} = \eta_{\text{CA}} \equiv 1 - \sqrt{\frac{T_L}{T_H}}.$$

- Agree with Curzon and Ahlborn's prediction [Am. J. Phys 43, 22 (1975)].

Overdamped limit

- In the overdamped limit $\kappa \gg \lambda$, $\mathcal{L} = \oint \sqrt{\frac{\kappa}{(\lambda_0 e^r)^2}\frac{1}{T}\left(Tdr - \frac{dT}{2}\right)^2}$.
- Choose the exponential paths with $\alpha = 2$ (zero length), and obtain

$$P_{\max}^{\text{over}} = \frac{\lambda_0^2 (T_H - T_L)^2 (r_2 - r_1)^2}{16\kappa T_L (e^{-r_1} - e^{-r_2})^2},$$

$$\eta_{\text{EMP}}^{\text{over}} = \frac{\eta_{\text{C}}}{2 + \left[\frac{1}{(r_2 - r_1)} - \frac{1}{2}\right]\eta_{\text{C}}}.$$

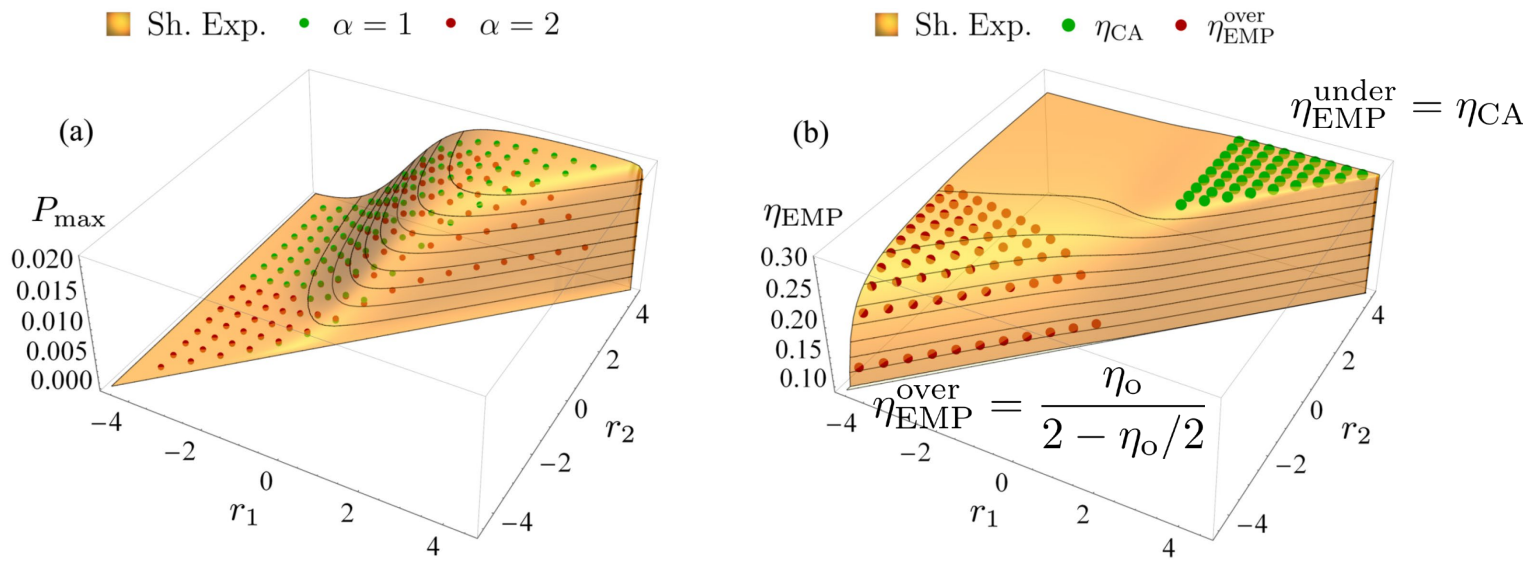
- The quasistatic efficiency is

$$\eta_{\text{o}}^{\text{over}} = \frac{\eta_{\text{C}}}{1 + \eta_{\text{C}}/[2(r_2 - r_1)]} < \eta_{\text{C}}.$$

Heat leakage from momentum degree of freedom

- EMP matches that in [T. Schmiedl and U. Seifert, Europhys. Lett. 81, 20003 (2007)].

$P_{\max}$ and η_{EMP} in general damped situation

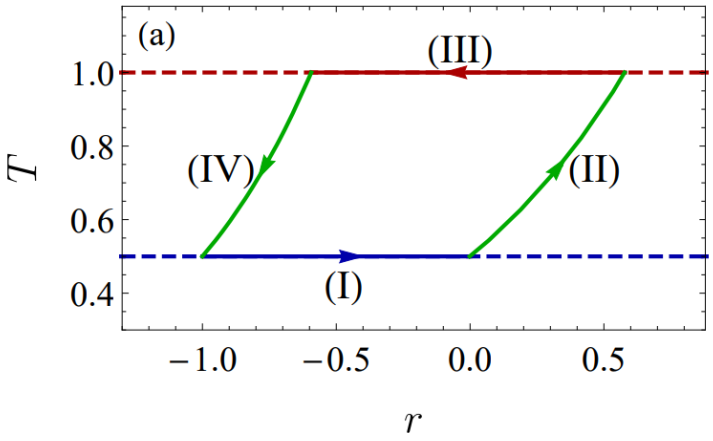


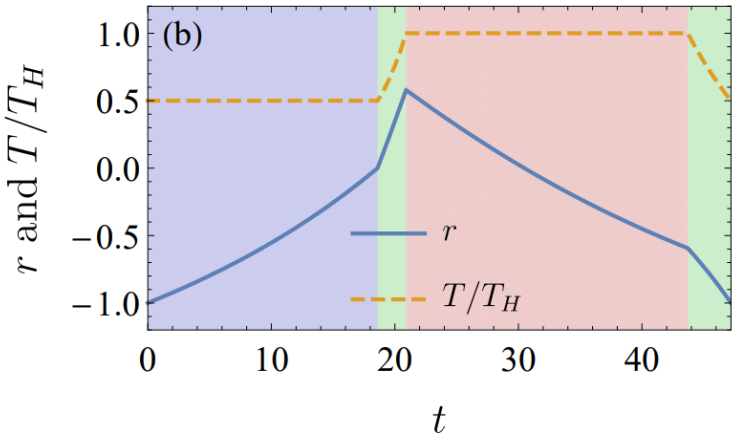
- $P_{\max}$ is the largest with the shortest exponential paths.
- Underdamped regime ($r \gg 0$): large $P_{\max}$, Curzon-Ahlborn's type.
- Overdamped regime ($r \ll 0$): small $P_{\max}$, Schmiedl-Seifert's type.

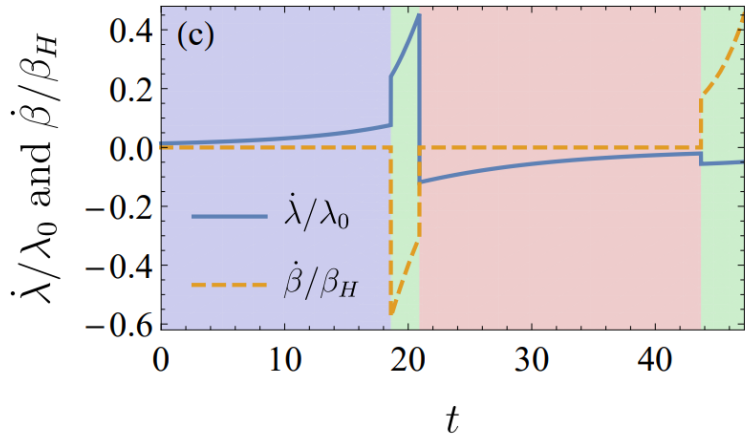
Control of the cycle

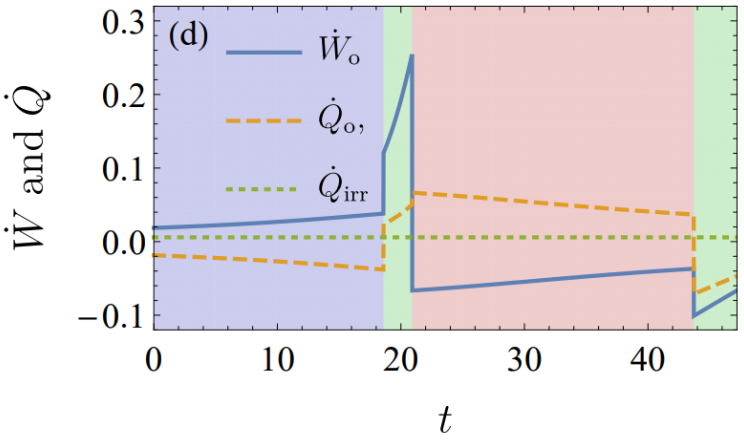
- The control of a typical cycle

Connecting processes consume time according to thermodynamic length



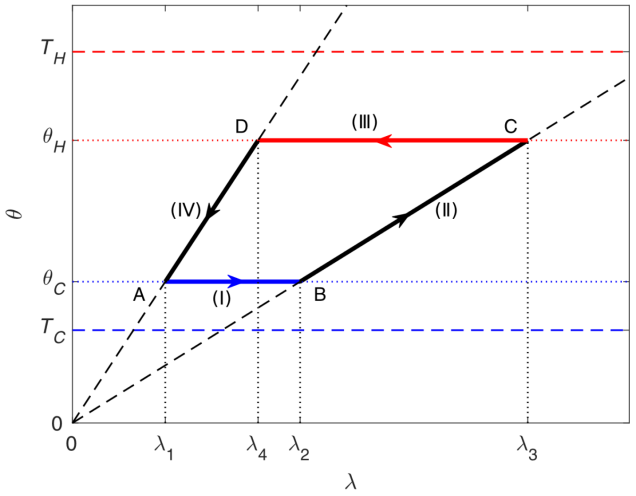


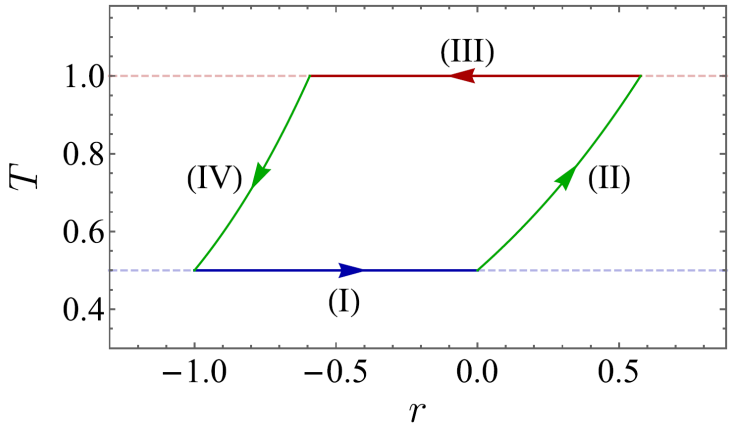




JFC, Phys. Rev. E 106, 054108 (2022)

Traditional optimization vs. Shortcut strategy





- Given protocol, solve cycle diagram.
- Irreversibility: deviation from quasi-static cycles.
- Thermodynamic length valid for slow driving processes.
- Optimal cycles are only solved in overdamped limit [T. Schmiedl and U. Seifert, Europhys. Lett. 81, 20003 (2007)]
- and underdamped limits [A. Dechant, N. Kiesel, and E. Lutz, Europhys. Lett. 119, 50003 (2017) and Y. H. Chen, JFC, Z. Fei, and H. T. Quan, Phys. Rev. E 106, 024105 (2022)].

- Given cycle diagram, solve $H_{\mathrm{a}}(t)$.
- Irreversibility: energy cost to implement $H_{\mathrm{a}}(t)$.
- Exact thermodynamic length for arbitrary finite-time processes.
- Geometric interpretation of the optimization to acquire large output power

$$P_{\max} = \frac{\mathcal{A}^2}{4\mathcal{L}^2}.$$

Summary and Outlook

- Shortcut strategy: a new way to construct and optimize the Brownian heat engines.
- Generalization to power-law potentials $U_{\rm o}(x,\lambda)=\frac{1}{2n}m\lambda^{n+1}x^{2n}$.
- Strict $1/\tau$ scaling of irreversible heat in shortcut processes.
- The performance of the optimal heat-engine cycle with shortcut strategy is characterized by $\mathcal{A}$ and $\mathcal{L}$.
- Better understanding of the optimal cycles in the underdamped and the overdamped limits.
- Extension to open quantum systems under external driving and also active systems.

JFC, Phys. Rev. E 106, 054108 (2022)





Thank You


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- Approximate shortcuts with momentum-independent auxiliary Hamiltonian see

  G. Li, C. P. Sun, H. Dong, Phys. Rev. E, 107, 014103 (2023)

- Lindblad master equation

$$\dot{\rho} = -i[H, \rho] + \sum_k \gamma_k (L_k \rho L_k^\dagger - \frac{1}{2} L_k^\dagger L_k \rho - \frac{1}{2} \rho L_k^\dagger L_k).$$